



TEMPLATE - The 11-Section Framework

How every topic in DSA Master is structured (Curriculum + Tier-Based Patterns)

Generated: 2025-12-26
Status: ☒ Complete with Tier Integration

Overview

Every topic follows this **consistent 11-section template**. This consistency helps your brain recognize patterns across different topics and deepens retention.

All topics (curriculum and tier-based patterns) use the same 11-section framework, with variations in length and depth based on topic type and tier level.



The Complete 11-Section Template

Section 0: Metadata Header

```
# [Topic Name]: [Subtitle]

## 📅 Metadata
**Topic:** [Topic Name]
**Week:** Week [N]
**Day:** Day [X] of 5
**Category:** [Category Name]
**Difficulty:** 🟢 Easy / 🟡 Medium / 🔴 Hard
**Status:** 🔍 In Study
**Last Reviewed:** –
**Confidence Level (1-5):** –/5
```

Purpose: Provides context and allows tracking progress.

Section 1: The Why — Engineering Motivation

```
## 1 The Why – Engineering Motivation

### Real-World Problem
[Describe the concrete problem this topic solves. Include specific examples.]

Example format:
- You have 1 billion records and need to search them in milliseconds
- You need to handle unknown incoming stream size efficiently
- You're building a kernel scheduler and need fast task selection
```

Design Problem Solved

- [Design goal 1]
- [Design goal 2]
- [Design goal 3]

Trade-offs Introduced

[What do we give up to get these benefits?]

Real System Usage

[Where does this appear in production?]

- Operating systems: [specific example]
- Databases: [specific example]
- Networks: [specific example]
- Graphics: [specific example]

Purpose: Ground learning in reality. Answer "why should I care?"

Length Guidelines:

- Curriculum Topics: 150-200 lines
- Tier 1 Patterns: 150-200 lines
- Tier 2 Patterns: 120-150 lines
- Tier 3 Patterns: 100-130 lines

Section 2: The What — Mental Model & Intuition

2 The What — Mental Model & Intuition

Core Analogy

[Explain using an everyday analogy. Make it vivid and memorable.]

Example: "Think of arrays as a row of numbered lockers..."

Visual Representation

[ASCII diagram showing how the concept looks visually]

Example:

[Show concrete visual]

Core Invariants

[List the fundamental properties that always hold]

Example:

- Invariant 1: [description]
- Invariant 2: [description]

```
- Invariant 3: [description]

### Key Concepts
[Define the core concepts simply]
```

Purpose: Build mental model before mechanics. Answer "what is this conceptually?"

Length Guidelines:

- Curriculum Topics: 100-150 lines
- Tier 1 Patterns: 100-150 lines
- Tier 2 Patterns: 80-120 lines
- Tier 3 Patterns: 70-100 lines

Section 3: The How — Mechanical Walkthrough

```
## 3 The How – Mechanical Walkthrough

### State/Data Structure
[Describe what state is maintained]

Example:
```

buffer: pointer to contiguous array size: number of elements capacity: maximum capacity

```
### Operation 1: [Name]
[Step-by-step mechanical walkthrough]

1. [First step]
2. [Second step]
3. [Third step]
Cost: [Time complexity]

### Operation 2: [Name]
[Similar step-by-step walkthrough]

### Memory Behavior
[How does this interact with memory, cache, pointers?]

### Edge Cases
[What can go wrong during operation?]
```

Purpose: Understand mechanics without code. Answer "how does it actually work?"

Length Guidelines:

- Curriculum Topics: 120-180 lines
- Tier 1 Patterns: 120-180 lines
- Tier 2 Patterns: 90-140 lines
- Tier 3 Patterns: 80-120 lines

Section 4: Visualization — Simulation & Examples

```
## 4 Visualization – Simulation & Examples

### Example 1: [Scenario]
[Show the initial state]
```

[ASCII diagram showing state]

```
[Walk through the execution]

Step 1: ...
Step 2: ...
Result: ...

### Example 2: [Different Scenario]
[Repeat with different example]

### Example 3: Edge Case
[Show an edge case]
```

Purpose: See concrete examples you can trace by hand. Answer "show me an example I can trace."

Length Guidelines:

- Curriculum Topics: 80-120 lines (3 examples)
- Tier 1 Patterns: 80-120 lines (3 examples)
- Tier 2 Patterns: 60-100 lines (2-3 examples)
- Tier 3 Patterns: 50-80 lines (2 examples)

Section 5: Critical Analysis — Performance & Robustness

```
## 5 Critical Analysis – Performance & Robustness

### Complexity Table

| Operation | Best | Average | Worst | Notes |
|-----|-----|-----|-----|-----|
```

```

| [Op 1] | O(?) | O(?) | O(?) | [Notes] |
| [Op 2] | O(?) | O(?) | O(?) | [Notes] |

### Memory Access Patterns
[How does this interact with cache?]
- Pattern A: ...
- Pattern B: ...

### Edge Cases & Failure Modes
- Failure 1: Description and what happens
- Failure 2: Description and what happens

### When Complexity Analysis Breaks Down
[What assumptions of Big-O don't hold in practice?]

```

Purpose: Understand costs, limitations, and hidden behaviors. Answer "what are the actual costs?"

Length Guidelines:

- Curriculum Topics: 100-150 lines
- Tier 1 Patterns: 100-150 lines
- Tier 2 Patterns: 80-120 lines
- Tier 3 Patterns: 60-100 lines

Section 6: Real System Integration

```

## 6 Real System Integration

### Operating Systems
[Where does this appear in the kernel?]

Example:
- Linux kernel: specific file and usage
- Process tables: how implemented
- Virtual memory: where this helps

### Databases
[How do databases use this?]

Example:
- B-trees use this for: ...
- Query optimization uses: ...
- Indexing strategy uses: ...

### Graphics & Gaming
[Usage in graphics engines]

Example:
- Vertex buffers: ...
- Rendering pipeline: ...

```

```

### Networking
[Network protocol usage]

Example:
- TCP/IP: ...
- Router: ...

### Compiler & Language
[How language runtimes use this]

Example:
- JVM uses: ...
- Python CPython uses: ...
- C++ STL uses: ...

```

Purpose: See this isn't abstract theory; it's everywhere. Answer "where is this in real code?"

Length Guidelines:

- Curriculum Topics: 80-120 lines
- Tier 1 Patterns: 80-120 lines
- Tier 2 Patterns: 60-100 lines
- Tier 3 Patterns: 50-80 lines

Section 7: Concept Crossovers

```

## 7 Concept Crossovers

### What It Builds On (Prerequisites)
- [Concept A]: How it's used here
- [Concept B]: How it's used here

### What Builds On It (Successors)
- [Concept C]: Uses this for ...
- [Concept D]: Uses this for ...

### Applications in Algorithms
- [Algorithm 1]: Uses this for ...
- [Algorithm 2]: Uses this for ...

### Combinations with Other Techniques
- Combined with [Technique A]: enables ...
- Combined with [Technique B]: enables ...

```

Purpose: See how this fits in the broader DSA landscape. Answer "how does this relate to everything else?"

Length Guidelines:

- Curriculum Topics: 60-100 lines
- Tier 1 Patterns: 60-100 lines
- Tier 2 Patterns: 50-80 lines
- Tier 3 Patterns: 40-70 lines

Section 8: Mathematical & Theoretical Perspective

8 Mathematical & Theoretical Perspective

Formal Definition

[Mathematical definition, if applicable]

Example:

An array A represents a function $A: \{0, \dots, n-1\} \rightarrow V$ with ...

Proof Sketch

[Why does this work? What's the fundamental reason?]

Example:

Claim: Binary search takes $O(\log n)$ comparisons.

Proof: Each iteration halves the search space. After k iterations, we have $n/2^k$ elements. When $n/2^k = 1$, we have $k = \log n$ steps.

Recurrence Relation

[If applicable, the recurrence and its solution]

Example:

$T(n) = T(n/2) + O(1) \rightarrow T(n) = O(\log n)$

Theoretical Models

[Cache complexity, I/O model, or other relevant theory]

Purpose: Provide mathematical rigor for those who think that way. Answer "formally, how does this work?"

Length Guidelines:

- Curriculum Topics: 60-100 lines
- Tier 1 Patterns: 60-100 lines
- Tier 2 Patterns: 50-80 lines
- Tier 3 Patterns: 40-70 lines

Section 9: Algorithmic Design Intuition

9 Algorithmic Design Intuition

When to Use This

[Decision criteria for choosing this technique]

1. [Condition 1]: Use this because ...
2. [Condition 2]: Use this because ...
3. [Condition 3]: Use this because ...

When NOT to Use This

[Anti-patterns and when to avoid]

Example:

- Don't use arrays if: frequent mid-insert/delete
- Use linked lists instead if: ...

Decision Framework

[Quick decision tree for when to choose]

Example:

Need O(1) indexing? → YES: Use arrays → NO: Consider linked lists

Trade-off Scenarios

[Real scenarios showing trade-offs]

Example:

- Scenario A: Array better because ...
- Scenario B: Linked list better because ...

Purpose: Develop judgment about when to use what. Answer "when should I use this?"

Length Guidelines:

- Curriculum Topics: 80-120 lines
- Tier 1 Patterns: 80-120 lines
- Tier 2 Patterns: 70-100 lines
- Tier 3 Patterns: 60-90 lines

Section 10: Knowledge Check — Socratic Reasoning

10 Knowledge Check – Socratic Reasoning

****Question 1: [Open-ended question that reveals understanding]****

Your reasoning:

- [Hint/consideration 1]
- [Hint/consideration 2]
- [Why this matters]

****Question 2: [Different angle]****

Your reasoning:

- [Hint/consideration 1]
- [Hint/consideration 2]

****Question 3: [Application to new context]****

Your reasoning:

- [Hint/consideration 1]
- [Hint/consideration 2]

****Question 4: [Edge case or misconception challenge]****

Your reasoning:

- [Hint/consideration 1]
- [Hint/consideration 2]

Purpose: Force active reasoning; reveal gaps in understanding. Answer "can I think deeply about this?"

Length Guidelines:

- Curriculum Topics: 100-150 lines (3-4 questions)
- Tier 1 Patterns: 100-150 lines (7 questions - 1 per day)
- Tier 2 Patterns: 80-120 lines (5-7 questions)
- Tier 3 Patterns: 70-100 lines (5 questions)

Section 11: Retention Hook — Memory Anchors

1 1 Retention Hook – Memory Anchors

One-Line Essence

> ****[Single sentence capturing the core idea]****

Example: "Arrays are ordered lockers: $O(1)$ indexing via arithmetic, $O(n)$ edits, best-case cache behavior."

Mnemonic Device

*****[Acronym or memorable phrase]*****

Explanation:

- ****[Letter]:**** [concept]
- ****[Letter]:**** [concept]
- ****[Letter]:**** [concept]

Geometric/Visual Cue

[Sketch or description of visual memory aid]

Cognitive Lenses

Lens	Insight
-----	-----

| **Computational** | [How CPU/RAM sees it] |

| **Psychological** | [Common misconceptions] |

| **Design** | [Trade-off perspective] |

| **Historical** | [How it evolved] |

Purpose: Build long-term memory with multiple retrieval cues. Answer "how do I remember this for years?"

Length Guidelines: 40-80 lines (All types and tiers)

 Length & Content Guidelines by Topic Type

Curriculum Topics (Weeks 1-4, 6-12, 14-16)

Total Length: 400-500 lines

Total Words: 3,000-4,000

Reading Time: 90 minutes

Section Breakdown:

- Section 1: 150-200 lines
- Section 2: 100-150 lines
- Section 3: 120-180 lines
- Section 4: 80-120 lines
- Section 5: 100-150 lines
- Section 6: 80-120 lines
- Section 7: 60-100 lines
- Section 8: 60-100 lines
- Section 9: 80-120 lines
- Section 10: 100-150 lines
- Section 11: 40-80 lines

Examples: 3 fully traced

Diagrams: 5-10

Questions: 3-4

Practice Problems: 10-15

Tier 1 Pattern Files (Week 4.5)

Importance: ESSENTIAL (70-80% interview coverage)

Total Length: 400-500 lines

Total Words: 6,000+ words

Reading Time: 90 minutes

File Naming: `Week_45_Day_0[1-4]_Combined.md`

Tier 1 Patterns:

1. Hash Map/Set (Day 1) - 70% usage

2. Monotonic Stack (Day 2) - 20-30% usage
3. Merge Operations (Day 3) - 30% usage
4. Partition (Day 4A) - 15% usage
5. Kadane's Algorithm (Day 4B) - 10% usage

Section Breakdown:

- Section 1: 150-200 lines (Why: Real-world urgency)
- Section 2: 100-150 lines (What: Pattern essence)
- Section 3: 120-180 lines (How: Algorithm steps)
- Section 4: 80-120 lines (Visualization: 3 examples)
- Section 5: 100-150 lines (Complexity: Critical for interview)
- Section 6: 80-120 lines (Real Systems: Databases, OS)
- Section 7: 60-100 lines (Crossovers: Where pattern applies)
- Section 8: 60-100 lines (Theory: Correctness proofs)
- Section 9: 80-120 lines (Decision: When vs other patterns)
- Section 10: 100-150 lines (Questions: 7 Socratic - 1 per section/day)
- Section 11: 40-80 lines (Memory: Pattern essence)

Examples: 3 fully traced with variations

Diagrams: 8-10

Questions: 7 (1 per day for Week 4.5)

Practice Problems: 5-7 minimum

Interview Usage %: Specified for each pattern

Support Files:

- [Week_45_Guidelines.md](#) - Navigation & structure guidance
- [Week_45_Questions.md](#) - All 28 Socratic questions (7 per pattern day)
- [Week_45_Checklist.md](#) - Daily & weekly progress tracking

Tier 2 Pattern Files (Weeks 5-5.5)

Importance: IMPORTANT (+10-12% interview coverage)

Total Length: 350-400 lines

Total Words: 4,500-5,500

Reading Time: 75 minutes

File Naming: [Week_5_Pattern_\[Name\].md](#) or [Week_55_Pattern_\[Name\].md](#)

Tier 2 Patterns:

1. Difference Array (Week 5) - 10-15% usage
2. In-Place Replacement (Week 5) - 8-12% usage
3. Deque Operations (Week 5.5) - 5-10% usage

Section Breakdown:

- Sections 1-11: Similar to Tier 1 but more concise
- Section 4: 60-100 lines (2-3 examples)

- Section 10: 80-120 lines (5-7 questions)

Examples: 2-3 with key variations

Diagrams: 6-8

Questions: 5-7

Practice Problems: 5 per pattern

Interview Usage %: Specified for each pattern

Tier 3 Pattern Files (Week 13+)

Importance: GOOD-TO-KNOW (+5-8% interview coverage)

Total Length: 300-350 lines

Total Words: 3,500-4,500

Reading Time: 60 minutes

File Naming: `Week_13_Pattern_[Name].md` or `Week_65_Pattern_[Name].md`

Tier 3 Patterns:

1. Fast & Slow Pointers Extended - 3-5% usage
2. Reverse & Two Pointers - 5-8% usage
3. Matrix Traversal - 3-5% usage
4. Conversion & Encoding - 2-3% usage

Section Breakdown:

- Similar to curriculum but slightly shorter
- Section 4: 50-80 lines (2 examples)
- Section 10: 70-100 lines (5 questions)

Examples: 2 with clear application

Diagrams: 5-7

Questions: 5

Practice Problems: 3-5 per pattern

Interview Usage %: Specified for each pattern

Complete File Naming Convention

Curriculum Topics

```
Week_[N]_Day_[X]_[Topic_Name].md
```

Examples:

```
Week_1_Day_1_RAM_Model_Pointers.md
```

```
Week_3_Day_1_Elementary_Sorts.md (NOT Week 3 Day 1 Sorting & Hashing title)
```

```
Week_5_Day_1_Binary_Tree_Anatomy.md (Week 5 is Trees & Heaps, NOT Hashing!)
```

```
Week_12_Day_5_System_Review_Integration.md
```

Tier 1 Patterns (Week 4.5)

Week_45_Day_0[1-4]_Combined.md

Examples:

Week_45_Day_01_Combined.md (Hash Map/Set)
Week_45_Day_02_Combined.md (Monotonic Stack)
Week_45_Day_03_Combined.md (Merge Operations)
Week_45_Day_04_Combined.md (Partition & Kadane's)

Support Files:

Week_45_Guidelines.md
Week_45_Questions.md
Week_45_Checklist.md

Tier 2 Patterns (Weeks 5-5.5)

Week_5_Pattern_[Name].md OR Week_55_Pattern_[Name].md

Examples:

Week_5_Pattern_Difference_Array.md
Week_5_Pattern_InPlace_Replacement.md
Week_55_Pattern_Deque_Operations.md

Tier 3 Patterns (Week 13+)

Week_13_Pattern_[Name].md OR Week_65_Pattern_[Name].md

Examples:

Week_13_Pattern_FastSlow_Pointers_Extended.md
Week_13_Pattern_Reverse_TwoPointers.md
Week_13_Pattern_Matrix_Traversal.md
Week_13_Pattern_Conversion_Encoding.md

How to Use This Template

For Creating Curriculum Topics

1. Copy this template
2. Replace [brackets] with actual content
3. Aim for 400-500 lines total
4. Include 3 fully traced examples
5. Make Socratic questions challenging but guiding
6. Ground all real systems sections in actual code

7. Include complexity tables
8. Follow all 11 sections exactly

Example: Week 3, Day 1: Elementary Sorts

- Why: Need efficient data organization
 - What: Bubble, insertion, selection sorts
 - How: Trace through sorting steps
 - Real systems: When used in production, Linux kernel
 - Length: 400-500 lines
-

For Creating Tier 1 Pattern Files (Week 4.5)

Importance Level: ESSENTIAL (70-80% interview coverage)

Steps:

1. Copy complete template
2. Expand Sections 1-11 to tier guidelines
3. Aim for 400-500 lines, 6000+ words
4. Include 3 fully traced examples
5. Create 7 Socratic questions (1 per day ideally)
6. Specify interview coverage % for each pattern
7. List 5-7 real practice problems
8. Create support files (Guidelines, Questions, Checklist)

Special Additions for Tier 1:

- **Interview Usage %:** Specify percentage of problems using this pattern
 - **Real Problems:** List 2-3 actual LeetCode/interview problems
 - **Why Essential:** Explain critical importance
 - **Patterns Chaining:** How to combine with other Tier 1 patterns
 - **Support Materials:** Create 3 additional files for guidance
-

For Creating Tier 2 Pattern Files (Weeks 5-5.5)

Importance Level: IMPORTANT (+10-12% interview coverage)

Steps:

1. Copy template
 2. Follow Tier 2 length guidelines (350-400 lines)
 3. Aim for 4,500-5,500 words
 4. Include 2-3 examples
 5. Create 5-7 Socratic questions
 6. Specify interview coverage %
 7. List 5 practice problems
-

For Creating Tier 3 Pattern Files (Week 13+)

Importance Level: GOOD-TO-KNOW (+5-8% interview coverage)

Steps:

1. Copy template
2. Follow Tier 3 length guidelines (300-350 lines)
3. Aim for 3,500-4,500 words
4. Include 2 examples
5. Create 5 Socratic questions
6. Specify interview coverage %
7. List 3-5 practice problems

☒ Quality Checklist

For All Topics (Curriculum & Patterns)

- ☒ Section 1 has 3+ real system examples
- ☒ Section 2 has memorable analogy
- ☒ Section 3 has mechanical walkthrough (no code)
- ☒ Section 4 has multiple traced examples
- ☒ Section 5 has complexity table
- ☒ Section 6 has 4+ system references
- ☒ Section 7 connects to other topics
- ☒ Section 8 provides rigor
- ☒ Section 9 has decision framework
- ☒ Section 10 has Socratic questions
- ☒ Section 11 has memorable one-liner
- ☒ Total lines within tier guidelines
- ☒ Multiple diagrams throughout
- ☒ Real code references
- ☒ Day/Week/Category properly labeled

For Tier Pattern Files (Additional)

- ☒ Pattern tier clearly identified (Tier 1, 2, or 3)
 - ☒ Interview usage percentage specified
 - ☒ Key insight clearly stated
 - ☒ Real problems listed (2+ for each tier)
 - ☒ Follows appropriate length guidelines
 - ☒ Can chain with other patterns
 - ☒ Clear when to use vs alternatives
 - ☒ Prerequisites listed
 - ☒ Follow-on concepts noted
 - ☒ Placement week/day specified
-

Example Topics Using This Template

Curriculum Topic Examples

Week 1, Day 1: RAM Model & Pointers

- Why: How do computers actually execute programs?
- What: Von Neumann model, memory hierarchy, pointers
- How: Execution trace of simple program
- Real systems: Process address space, virtual memory
- Length: 400-500 lines

Week 3, Day 1: Elementary Sorts

- Why: Understand sorting fundamentals
- What: Bubble, insertion, selection sorts
- How: Trace through sorting steps
- Real systems: When used in production
- Length: 400-500 lines

Week 5, Day 1: Binary Tree Anatomy (DIFFERENT from Week 3!)

- Why: Understand hierarchical data structures
- What: Tree structure and properties
- How: Tree navigation and traversal prep
- Real systems: DOM, file systems, databases
- Length: 400-500 lines

Tier Pattern Examples

Week 4.5, Day 1: Hash Map/Set (Tier 1)

- Interview Usage: 70% of problems
- Essential for: Two sum, anagrams, frequency counting
- Length: 400-500 lines, 6000+ words
- Questions: 7 Socratic
- Support: Guidelines, Questions (28 total), Checklist

Week 5.5: Difference Array (Tier 2)

- Interview Usage: 10-15% of range problems
- Essential for: Range updates, efficient queries
- Length: 350-400 lines, 4500-5500 words
- Questions: 5-7 Socratic
- Practice: 5 problems

Week 13+: Matrix Traversal (Tier 3)

- Interview Usage: 3-5% of matrix problems
- Good for: 2D array navigation
- Length: 300-350 lines, 3500-4500 words

- Questions: 5 Socratic
- Practice: 3-5 problems

Section 12: Additional Sections

Include in every topic file:

```
##  Cognitive & Meta Layers

[Optional: Additional cognitive frameworks]

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##  Revision & Spaced Repetition

Track your understanding over time:

| Review Date | Confidence (1-5) | Strengths | Areas to Deepen | Next Review |
|---|---|---|---|---|
| 2025-12-26 | - | - | - | 2025-12-28 |
| | | | | |

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##  Reference Pointers

### Textbooks
- [Book title]: [Relevant chapter/section]

### Online Resources
- [Website]: [Topic coverage]

### Real System Code
- [Project]: [File/function to study]

### Personal Insights & Notes
[Space for you to write discoveries]

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##  Navigation

**← [Previous topic]**
**→ [Next topic]**
**↑ [Back to Week [N] Summary]**
** [Back to Master Prompt]**
```

Implementation Workflow

Step 1: Start with Curriculum Topics (Weeks 1-4)

- Use template with 400-500 lines
- Create 5 topics per week
- Include all 11 sections
- Use provided line count guidelines

Step 2: Create Tier 1 Patterns (Week 4.5)

- Use template with 400-500 lines
- Create 5 pattern files (5 days + support)
- Follow Tier 1 guidelines
- Create support files (Guidelines, Questions, Checklist)
- Specify interview usage % for each

Step 3: Continue Curriculum (Weeks 5-12)

- Same as Step 1
- After Week 5, add Tier 2 patterns (Week 5.5)

Step 4: Create Tier 2 Patterns (Week 5.5)

- Use template with 350-400 lines
- Create 3 pattern files
- Follow Tier 2 guidelines

Step 5: Complete Curriculum (Weeks 6-12)

- Continue with 400-500 line template

Step 6: Create Tier 3 Patterns (Week 13+)

- Use template with 300-350 lines
- Create 4 pattern files
- Follow Tier 3 guidelines

 Quick Reference Table

Type	Length	Words	Time	Examples	Questions	Files
Curriculum	400-500	3-4K	90 min	3	3-4	1
Tier 1	400-500	6K+	90 min	3	7	5+3 support
Tier 2	350-400	4.5-5.5K	75 min	2-3	5-7	1 each
Tier 3	300-350	3.5-4.5K	60 min	2	5	1 each

 Key Principles

1. **Consistent Structure** - All 11 sections in every file
 2. **Real Systems First** - Ground theory in practice
 3. **Socratic Reasoning** - Force active thinking
 4. **Mental Models** - Teach understanding, not facts
 5. **Tier Clarity** - Specify interview coverage %
 6. **Support Materials** - Provide guidance & tracking
 7. **Length Flexibility** - Adjust by type & tier
 8. **Multiple Examples** - Trace by hand, see patterns
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Version: 2.0

Status: ☒ Complete with Tier Integration

Last Updated: 2025-12-26

Compatible With: Master Prompt v2.1